Patrick Pataky

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Master's student specializing in Computer Architecture and Digital Design. Experienced in RTL development (SystemVerilog/Verilog), FPGA-based hardware design and verification, and scripting for design automation. Looking for an RTL Design internship for Summer 2026.

Education

- **Ecole polytechnique fédérale de Lausanne (EPFL)** *2024 (In Progress)*
  Master in Computer Science (with Cyber Security Specialization)
- **Ecole polytechnique fédérale de Lausanne (EPFL)** *2021 - 2024*
  Bachelor in Computer Science

Technical Skills and Interests

Languages: SystemVerilog, Verilog, VHDL, C/C++, Python, CUDA, Java, Scala, Bash, ARM, RISC-V, x86
Tools & Frameworks: Vivado, QuestaSim, FPGA (Xilinx/Intel), Quartus, Yosys, Linux, Git, GDB, TCL Scripting
Areas of Interest: GPU Architecture, RTL Design & Microarchitecture, ASIC & FPGA Design, Digital Signal Processing, Computer Architecture, Object-Oriented Programming, Hardware-Software Co-Design
Languages Spoken: English (fluent), French (native), Slovak (basic)

Projects

- **X-HEEP RISC-V MCU: FPGA Verification and Flash Cache Design** *Spring 2026 (In Progress)*
  Semester Project at Embedded Systems Laboratory (ESL), EPFL
  - **Hardware Verification:** Verifying a Flash Controller on a Xilinx FPGA for the open-source X-HEEP RISC-V MCU, utilizing Integrated Logic Analyzers (ILA) to debug hardware and software issues across I/O operations.
  - **RTL Design & Implementation:** Designing a Write-Back Cache in SystemVerilog (holding 1 to 4 sectors) to optimize memory access latency.
  - **Testbench Development:** Developing comprehensive C test applications to stress-test the hardware controller and identify functional bugs. Usage of TCL scripts to automate simulations in QuestaSim.
  - **Keywords:** SystemVerilog, RTL Design, FPGA Verification, C/C++, Hardware Debugging (ILA), Testbenches, QuestaSim, TCL, Vivado, Performance Optimization, Hardware-Software Co-Design
- **FPGA Snake Game Hardware Accelerator** *Fall 2024*
  Course project from Embedded system design (CS-476)
  - **RTL Design & Implementation:** Developed a custom hardware accelerator from scratch on the Gecko 5 Education Board FPGA to offload graphical rendering from the CPU.
  - **Analytical Problem-Solving:** Achieved a 50x speedup over the software implementation by profiling C code to identify critical computational paths and designing optimized RTL on the FPGA.
  - **Keywords:** Verilog, C/C++, RTL Design & Implementation, Yosys, FPGA, GPU-style Hardware Accelerator, Performance Analysis, DSP
- **Bachelor Project: Remote Physical Attacks on Multi-Tenant FPGA Partial Reconfiguration** *Spring 2024*
  Semester project at PARSA, EPFL
  - **Conducted remote side-channel power analysis** using voltage routing delay sensors in a multi-tenant FPGA environment to capture power consumption traces during reconfiguration, depending on the attacker/victim placement.
  - **Keywords:** FPGA, Xilinx Zynq UltraScale, Dynamic Partial Reconfiguration, Hardware Verification, Python, Data Analysis, Vivado, ASIC/FPGA Platforms, TCL Scripting

Experience

- **Teaching Assistant** *Fall 2023 (In Progress)*
  EPFL, Lausanne (Switzerland)
  - **Courses**: CS-173 (Fundamentals of Digital Systems), CS-200 (Computer Architecture), CS-302 (Parallelism and Concurrency in Software), CS-470 (Advanced Computer Architecture)
  - **Delivered presentations and co-created course material**, including assignments and exams.
  - **Mentored 40-50 students weekly** providing guidance on digital logic, processor design, and programming labs.
  - **Debugged student implementations in real-time** for key projects, including designing and testing a simple in-order multicycle CPU from scratch in Verilog and writing a cycle-accurate C simulator of an out-of-order CPU.
- **IT Developer / Infrastructure Developer** *August 2023-September 2023*
  Abalance SA, Neuchâtel (Switzerland)
  - **Server installation**: Installed and configured Linux servers for a 10-person environment, and reduced manual setup time by 90% by automating repeatable steps with shell scripts.
  - **Maintenance**: Performed regular maintenance tasks to ensure updates across 3 servers.